

$$x_j = j\Delta x, \quad t_n = n\Delta t,$$

$$r = \frac{\kappa\Delta t}{(\Delta x)^2},$$

$$u_j^{n+1} = u_j^n + r(u_{j-1}^n - 2u_j^n + u_{j+1}^n),$$

$$0 \leq r \leq \frac{1}{2} \qquad \text{(explicit stability),}$$

$$u_0^n = u_N^n = 0, \quad \Delta x = L/N,$$

$$\text{error} = O(\Delta t + \Delta x^2) \qquad \text{for smooth solutions.}$$